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100G+ Optical Transport network technology

100G超光トランスポートネットワーク技術の最新動向



November 6th 2014

Taichi Kogure
Senior Systems Engineer,
Core Systems & Network Engineering

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Agenda

MPLS JAPAN

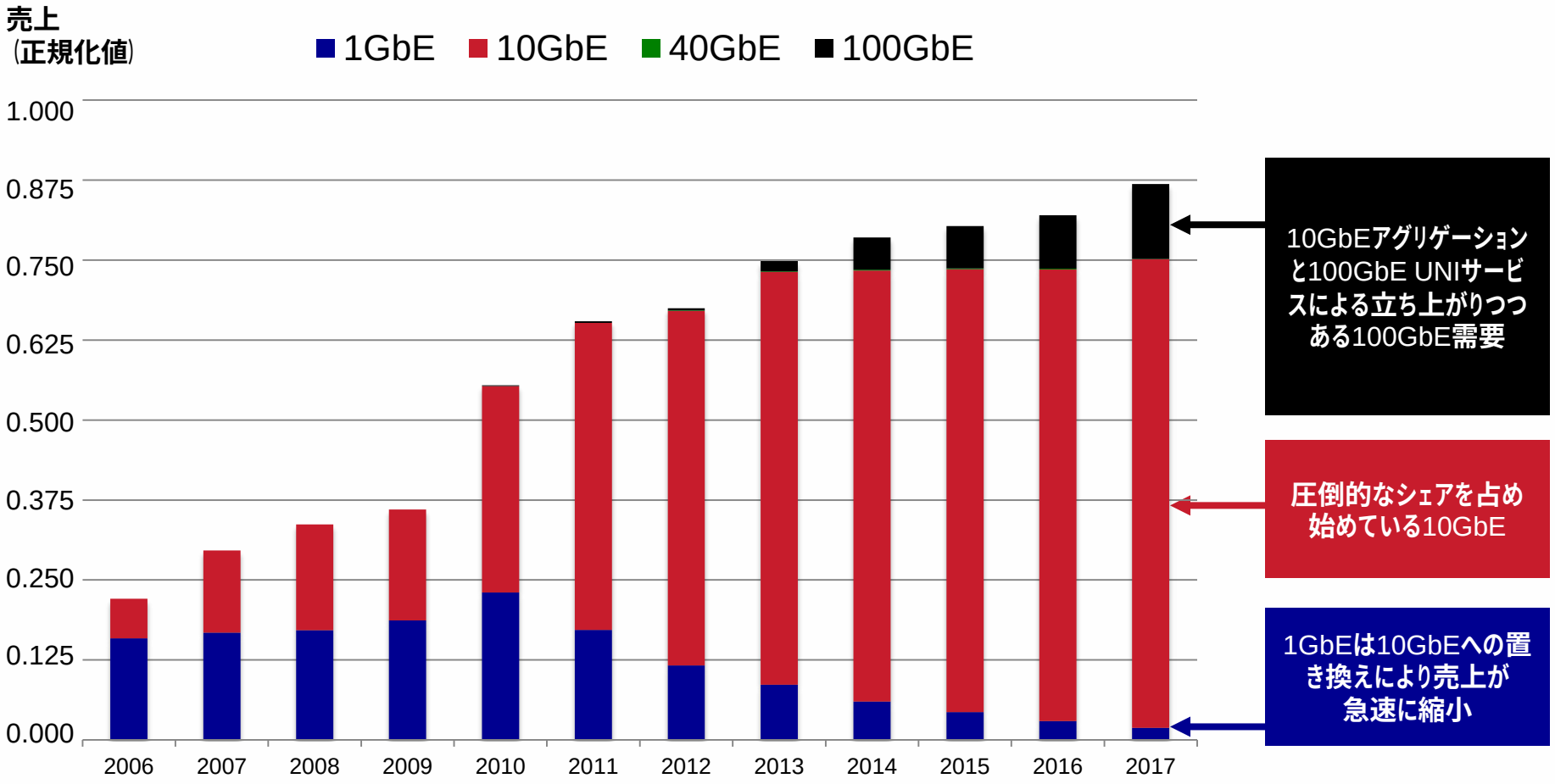
- Introduction
- Ciena WaveLogic3 Coherent DSP
- Submarine Feature
- Metro Feature
- What's next?

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イーサネット装置における売上高の推移

10GbEは新しい1GbE、100GbEは新しい10GbE



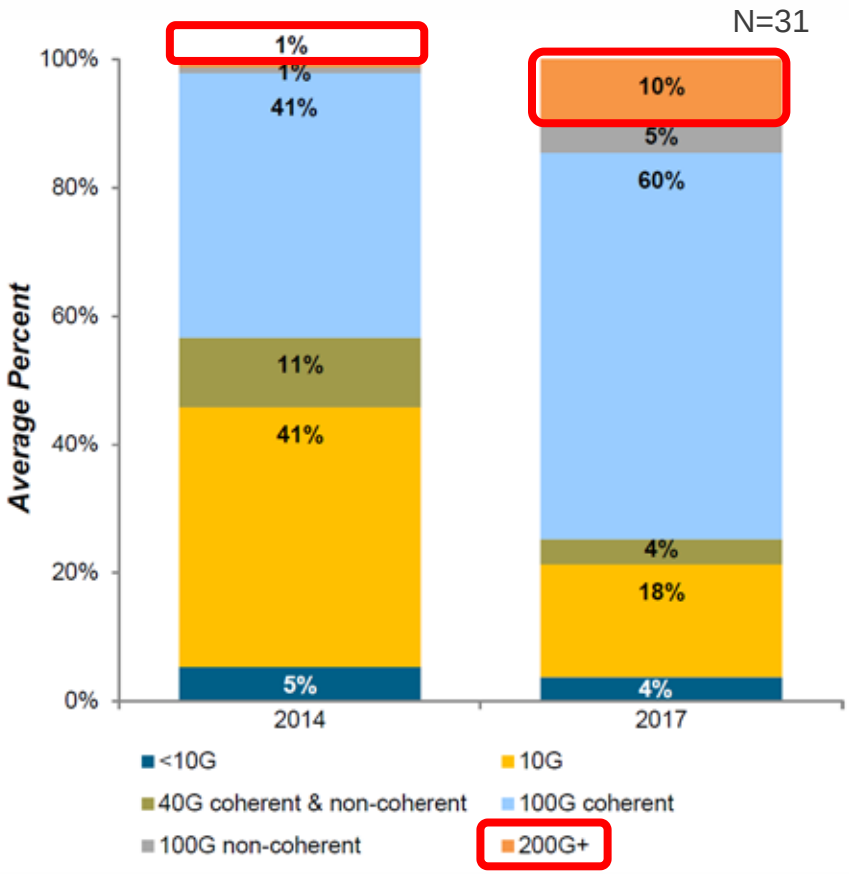
出典:「Service Provider Ports & Switches」2013年11月 (Infonetics)

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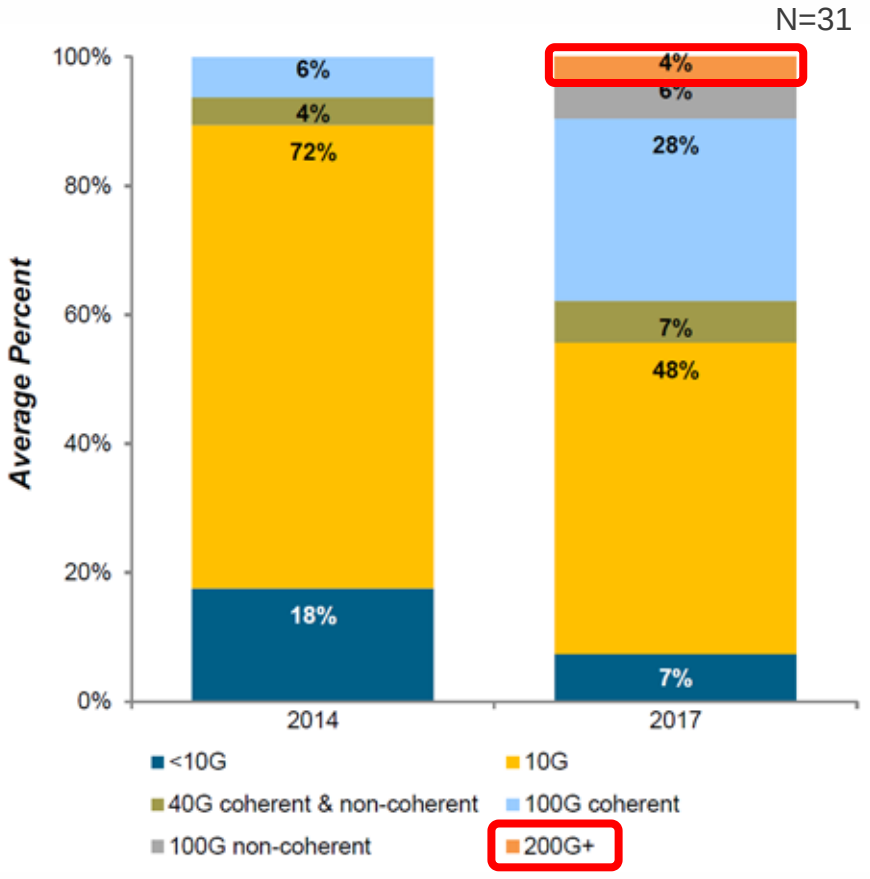
コア・メトロ領域における100G+の導入計画

200G, 400G, 1Tは現実に

Speeds and Modulations comparison 2014 versus 2017



New Core deployment



New Metro deployment

Source: Infonetics, “100G+ and ROADM Strategies, Global Service Provider Survey,” September 18, 2014

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- Introduction
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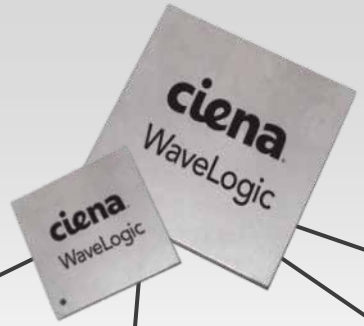
WaveLogic™ 3

Programmable Hardware Enabling Broad Solution Set

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SINGLE CHIPSET	MULTIPLE APPLICATIONS	FASTER SERVICE ACTIVATION

Debut on Sep, 2012

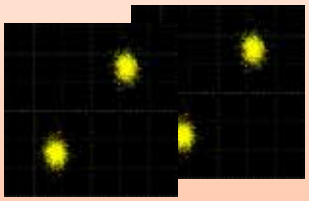


WaveLogic 3 Flex2

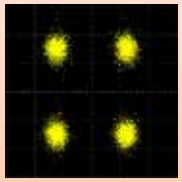
Released on Dec, 2012

WaveLogic 3 Flex3

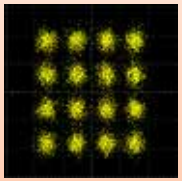
Shipping this Quarter



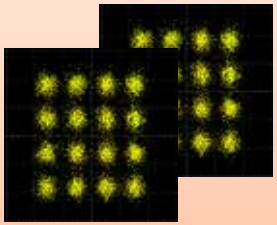
DC DP BPSK
Trans-Pacific
100Gb/s



DP QPSK
Ultra Long Haul
100Gb/s



DP 16QAM
Regional
200Gb/s



DC DP 16QAM
Regional
400Gb/s

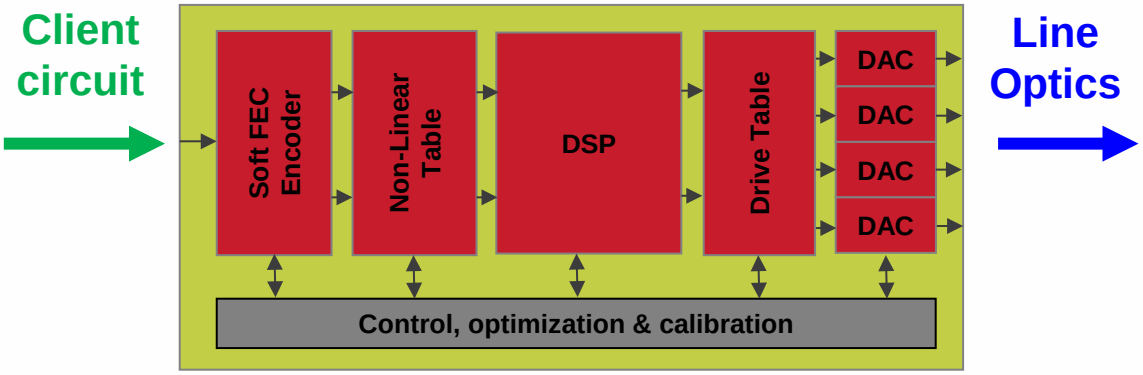
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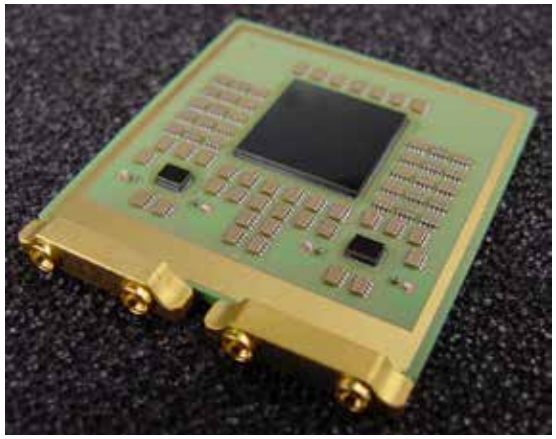
WaveLogic™ 3

DSP Chipset features

TX DSP

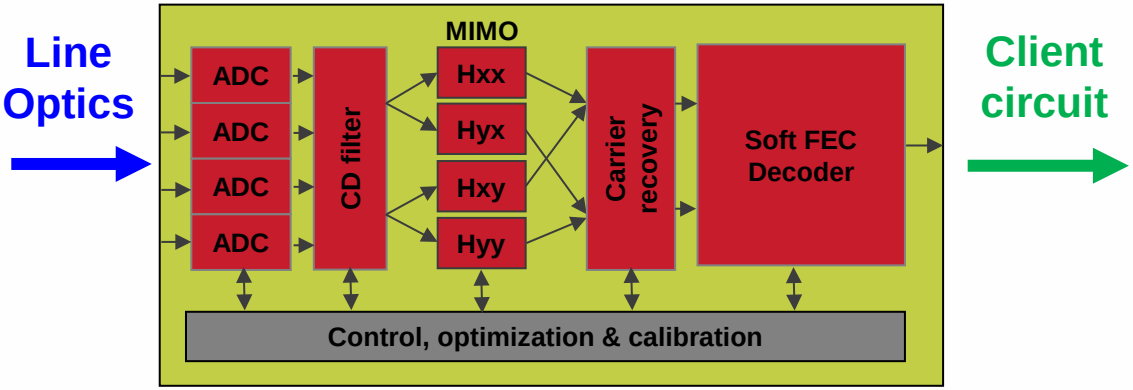


~ 22M Gates (DSP: 20M, DAC: 2M)

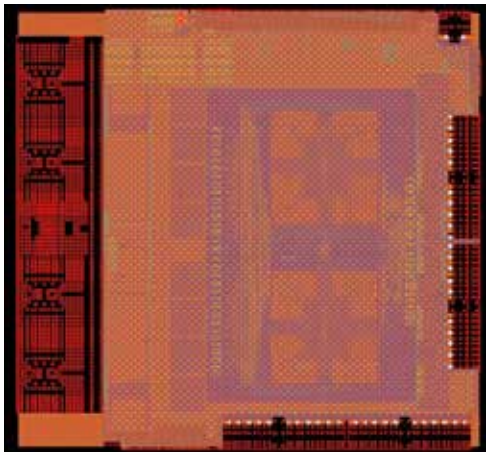


TX module

RX DSP



150M Gates (32 nm CMOS, 70 Tops/s, 3.7 km wire)



RX DSP chip layout

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WaveLogic™ 3 Flexible Transceiver

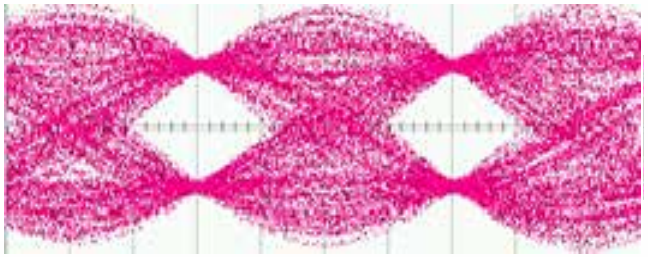


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WaveLogic™ 3 performance outlook

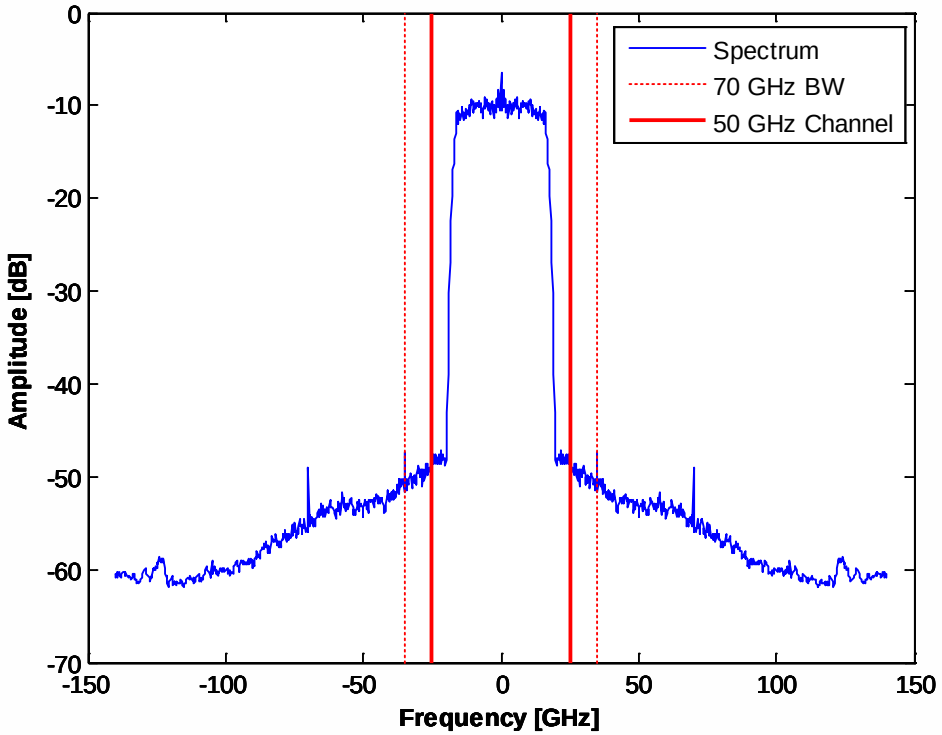
Spectral shaping enabling

- Minimize spectral utilization for increasing spectral efficiency
- DAC technology enables this functionality



Time Domain





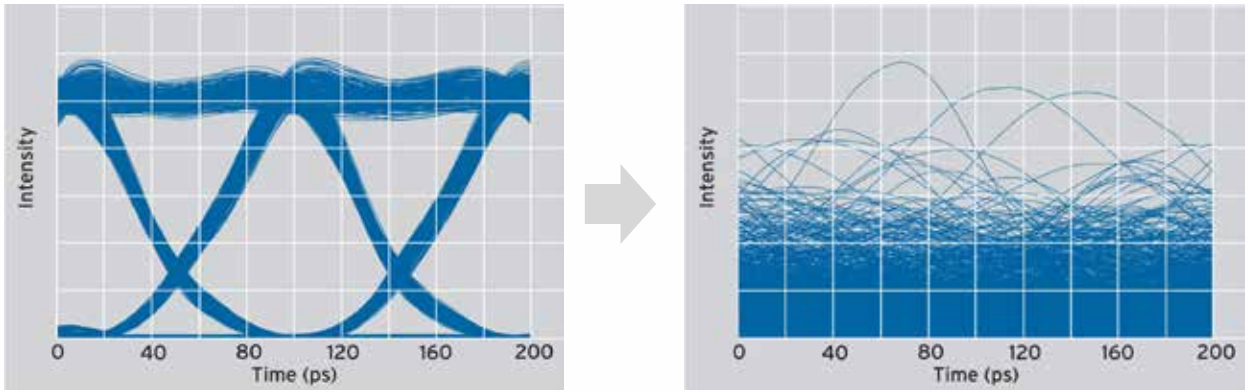
Frequency Domain

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WaveLogic 3 performance outlook

Linear distortion tolerance

Chromatic Dispersion tolerance

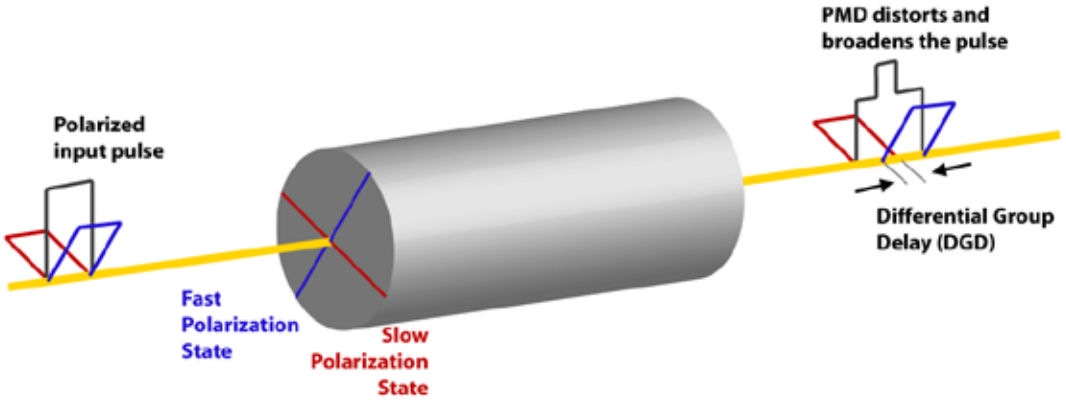


Transmitted

After Dispersion

100G Flex3
~300,000 ps/nm

Polarization Mode Dispersion tolerance



100G Flex3
150 ps (mean)

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WaveLogic 3 performance outlook

Flexible Latency Soft FEC

	100G	400G
Q-FEC	4 - 35 ms	2 - 18 ms
TX + RX Including Q-FEC	8 - 42 ms	5 - 25 ms



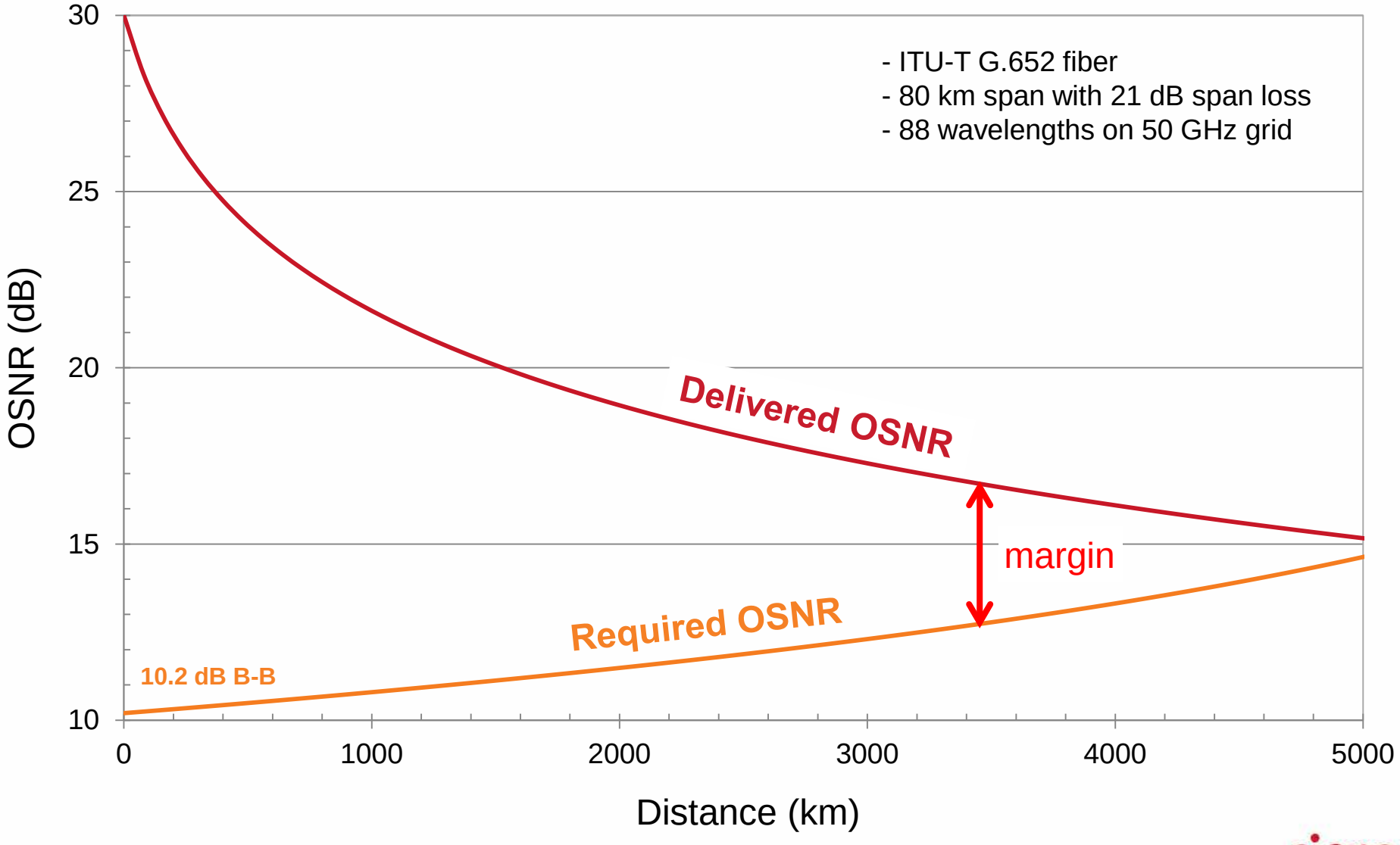
Q-FEC is flexible

- Latency is provisionable (out of service)
- Allows tradeoff between performance and latency

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WaveLogic 3 performance outlook

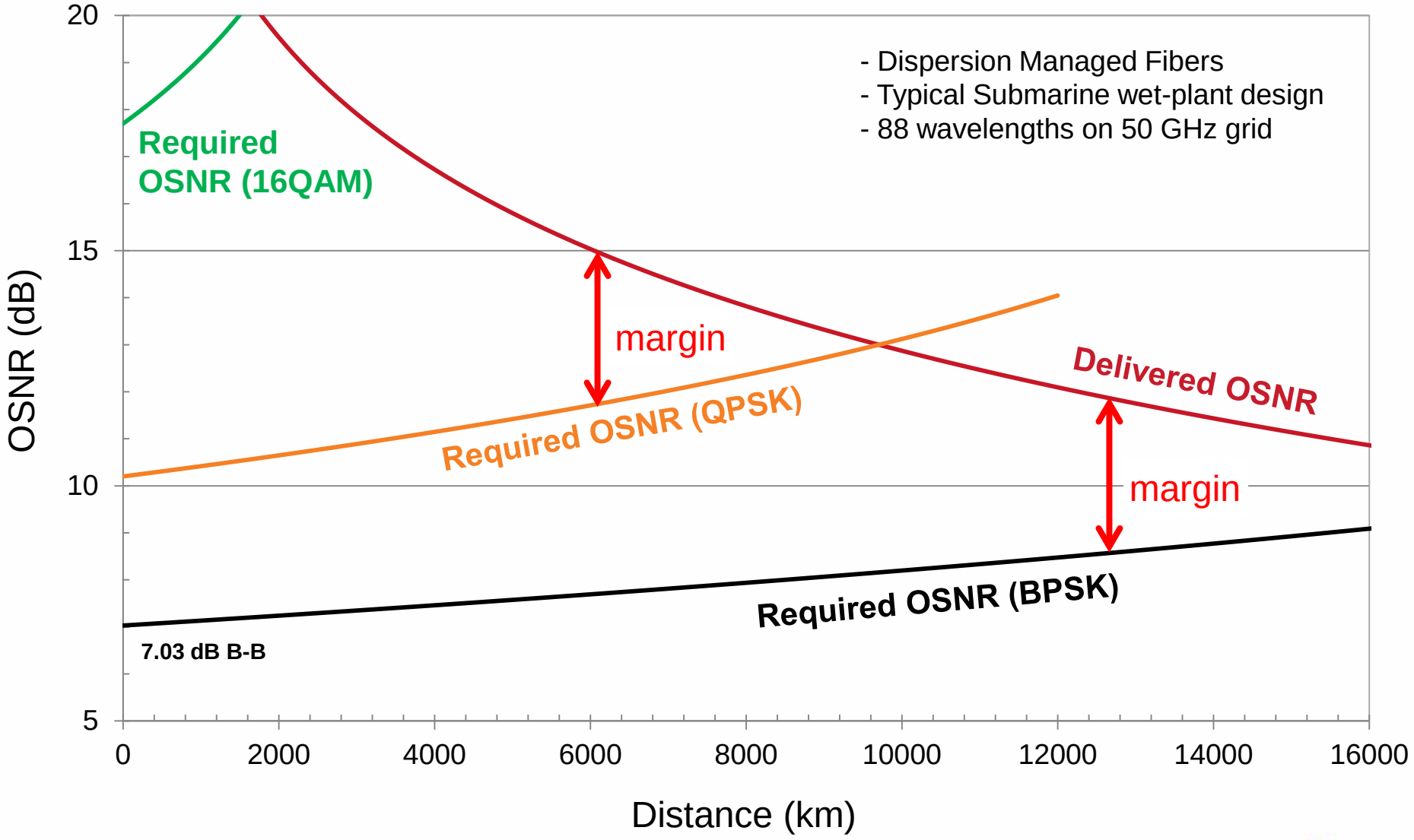
Typical Reach with 100G DP-QPSK on Terrestrial link



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WaveLogic 3 performance outlook

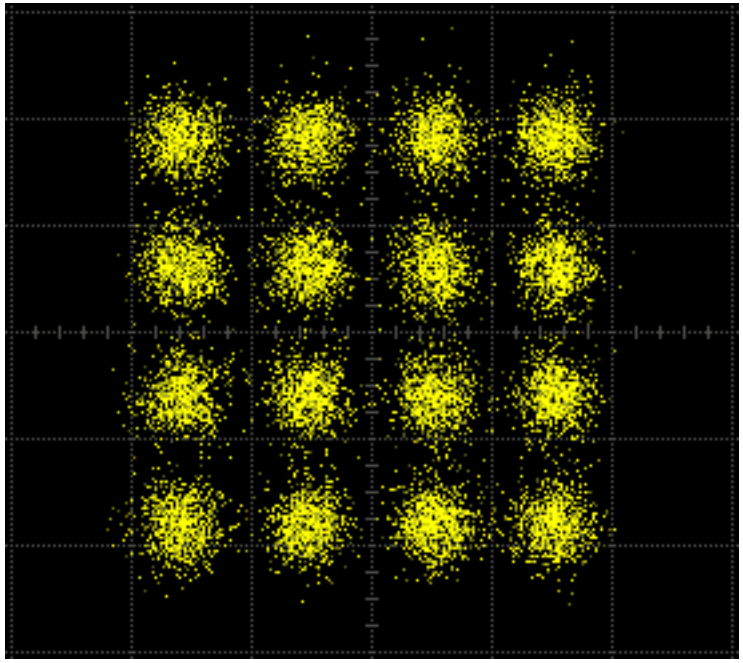
Typical Reach on Legacy Submarine link (DMF)



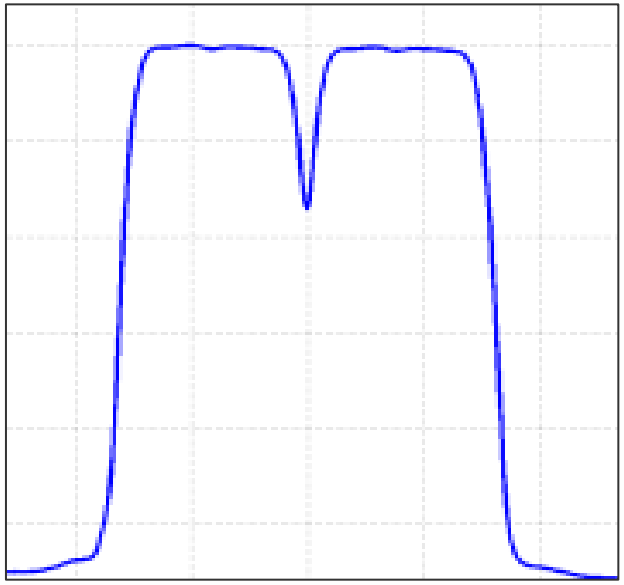
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WaveLogic 3 performance outlook

Typical Reach on 100G+ modulation



Constellation per carrier



Spectrum (Dual carriers)

Example Shown:

- 21 dB per span
- Hybrid Raman/EDFA amplifier
- 19 Tb/s

Fiber Type	Spans	Reach
Standard SMF	16	1,275 km
SMF-28 ULL	14	1,660 km
EX2000	17	2,000 km

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Trialing new technologies over current networks

200G/400G/800G Customer Field Trials

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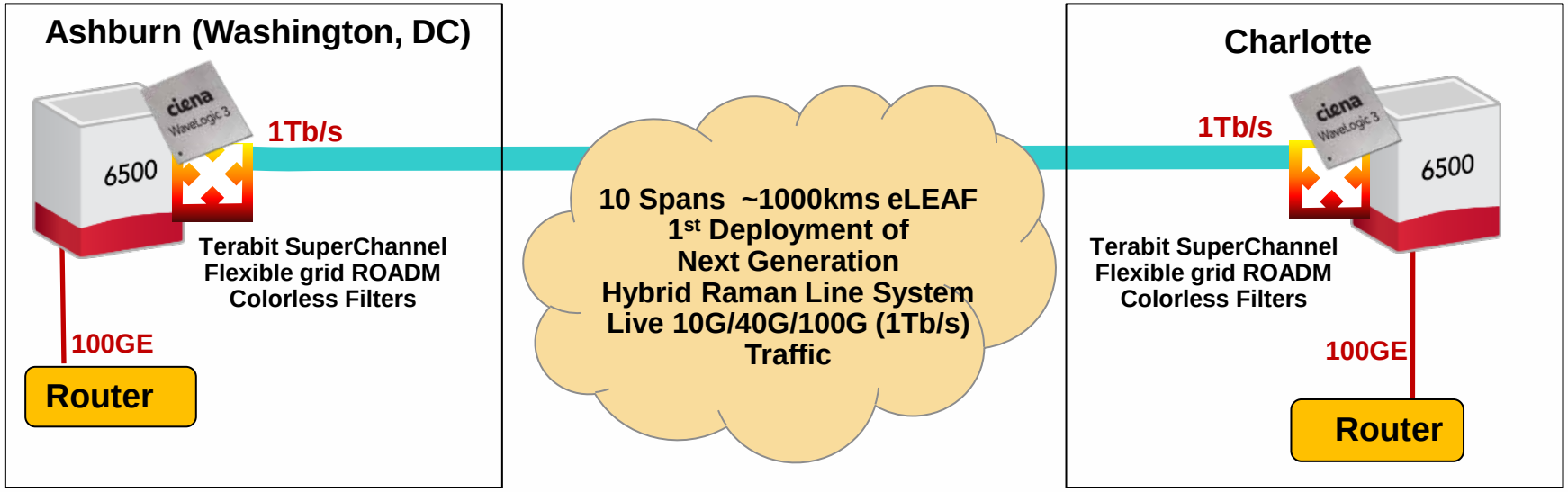
VERIZON	**Trial:** 200G carried alongside live customer traffic for one month period **Result:** Doubling of spectral efficiency with minimal incremental cost
SPRINT	**Trial:** 400G carried alongside live customer traffic **Result:** Existing network ready for 400GE Sprint Ethernet Wave Services
BRITISH TELECOM	**Trial:** 100G, 200G, 400G, 800G SuperChannel error-free operation **Result:** 800G SuperChannel on fiber that could not support 10G
JAPAN US CABLE NETWORK	**Trial:** 200G along submarine segment off the US West Coast **Result:** Doubling of capacity with 16QAM viable for existing submarine systems

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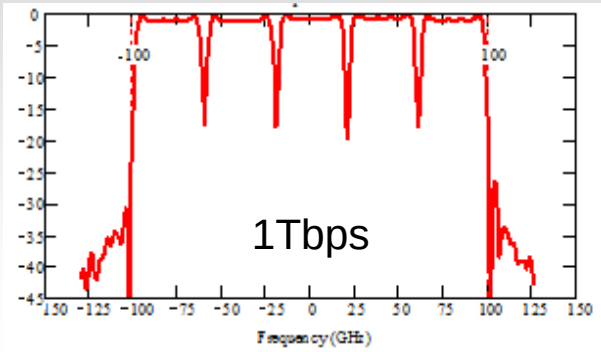
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Comcast 1Tb Field Trial

2.5x Increase in Spectral Efficiency



- Live Comcast Traffic over a single 1Tb/s SuperChannel wavelength
  - 5x16QAM 200GHz SuperChannel, operating on a next gen line system comprised of smart Raman, programmable EDFA & flexible grid ROADM
  - 1Tb/s, 200GHz SuperChannel operates alongside 10G, 40G, and 100G wavelengths running production traffic
  - 10 spans, eLEAF fiber, ~ 1000 km !



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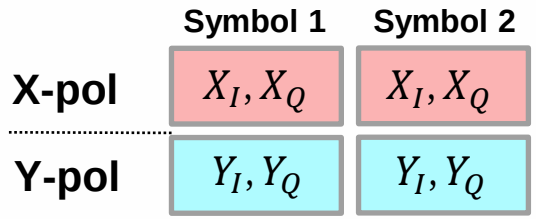
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- Metro Feature
- What's next?

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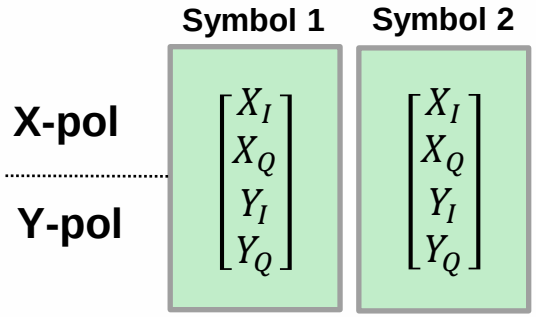
Multidimensional modulation formats

Standard modulation:
DP-BPSK, DP-QPSK, DP-16QAM, etc.



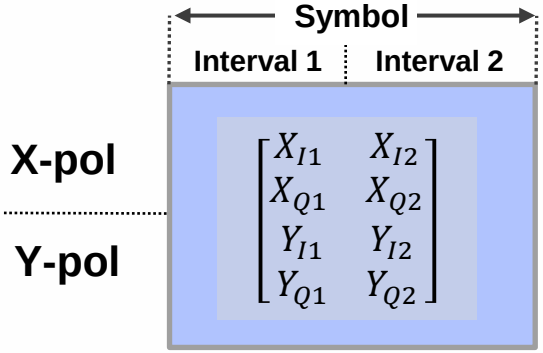
Bits encoded/decoded independently on each dimension.

4-Dimensional modulation:
Pol. Switched -QPSK, etc.



Symbols encoded across 4 optical dimensions. Increases Euclidean distance.

8-Dimensional modulation



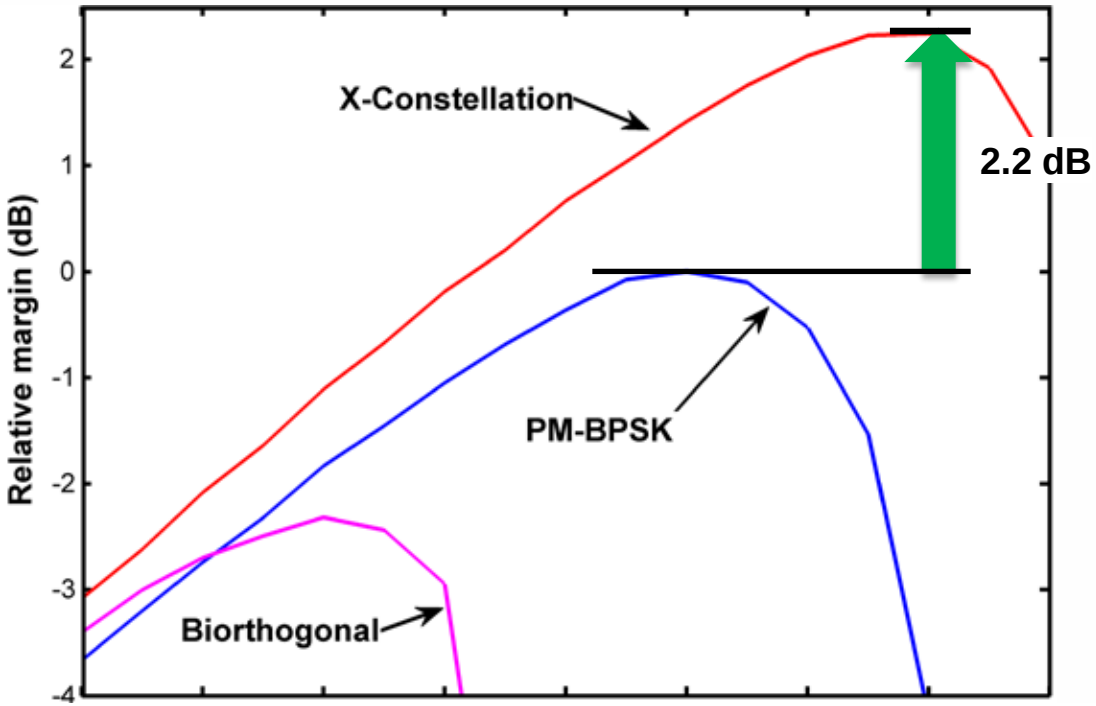
Symbols encoded across the 4 optical dimensions of two signaling intervals.

Symbols encoded across the additional dimension of time to realize improved linear and nonlinear performance.

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8D modulation Simulation Results

<Simulation Conditions>
- 5 Channel propagation
- 50 GHz channel spacing
- 20 x 80 km spans
  90% compensated ELEAF
- 28 GSymb/s
- 42 GHz Optical Filter



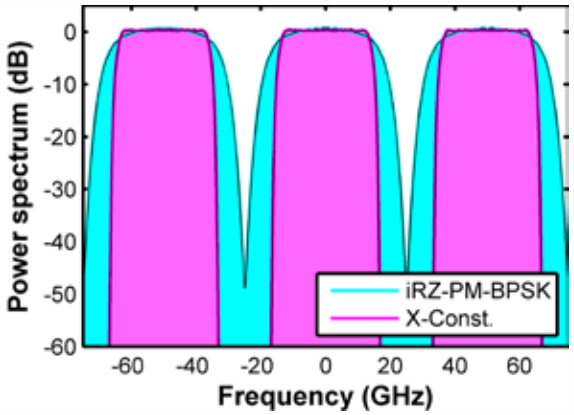
- X-Constellation rotated to:
  - → Ensures all time slots have the same symbol energy
  - → Balance the polarization stokes vectors for each pair of time slots.
- 2.2 dB improvement in Net System Margin relative to 'best case' PM-BPSK
- 4.5 dB better than un-rotated bi-orthogonal with same b2b performance.

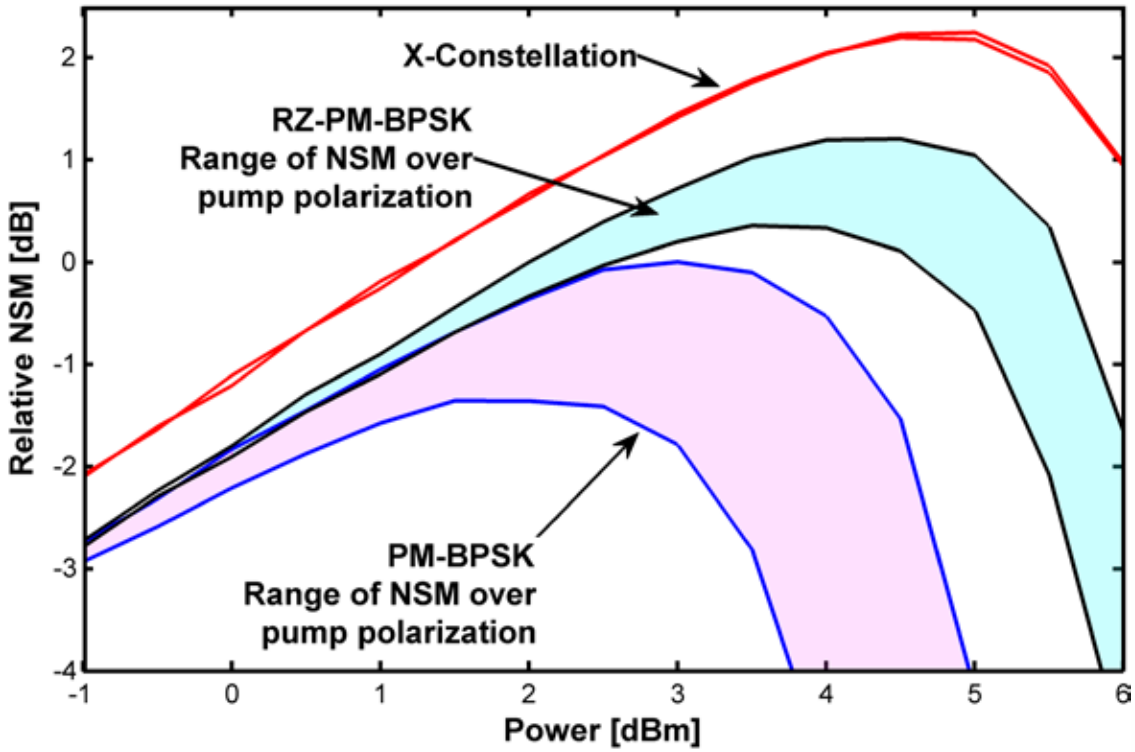
A. D. Shiner, et al., "Demonstration of an 8-dimensional modulation format with reduced inter-channel nonlinearities in a polarization multiplexed coherent system," Opt. Exp., Vol. 22, No. 17, 2014

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Polarization Sensitivity Comparison (Simulation)

<Simulation Conditions>
- 5 Channel propagation
- 50 GHz channel spacing
- 20 x 80 km spans
  90% compensated ELEAF
- 28 GSymb/s
- 42 GHz Optical Filter

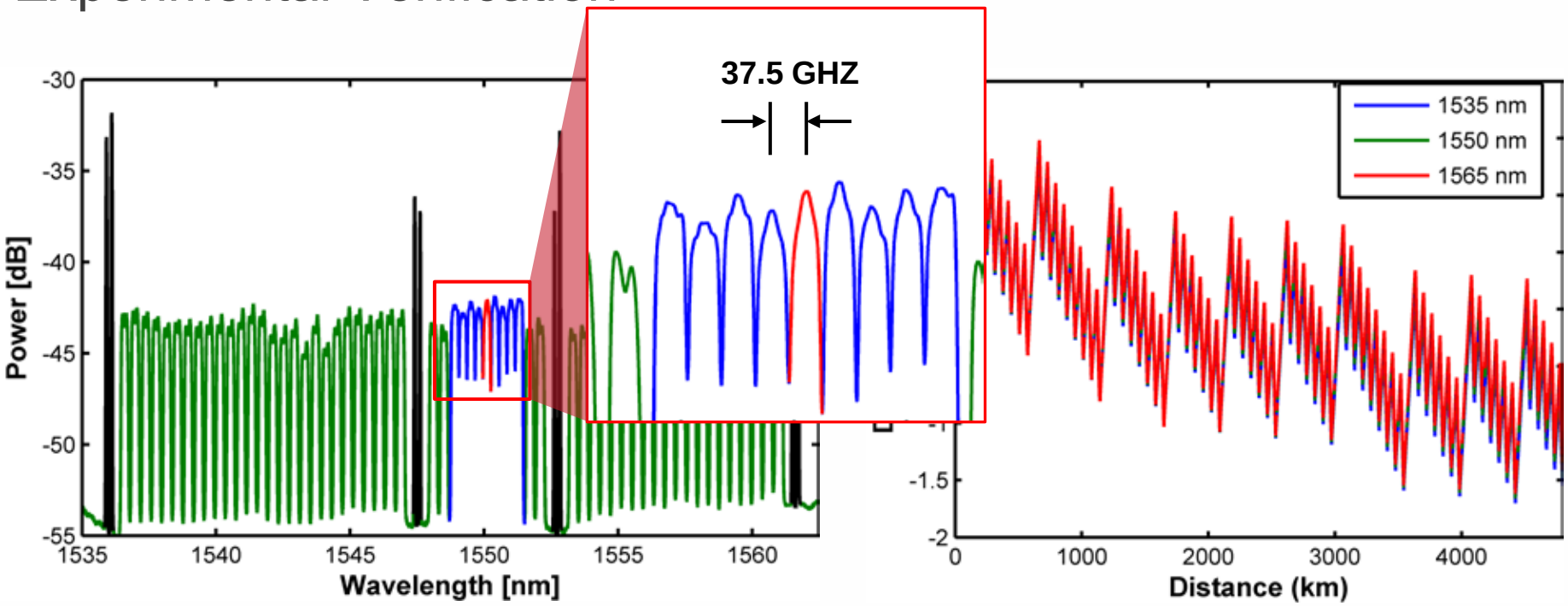




- PM-BPSK maximum NSM varies by 1.4 dB depending on neighboring channel polarization state.
- Symmetries in the X-constellation greatly reduce sensitivity to the polarization of the neighboring channels.

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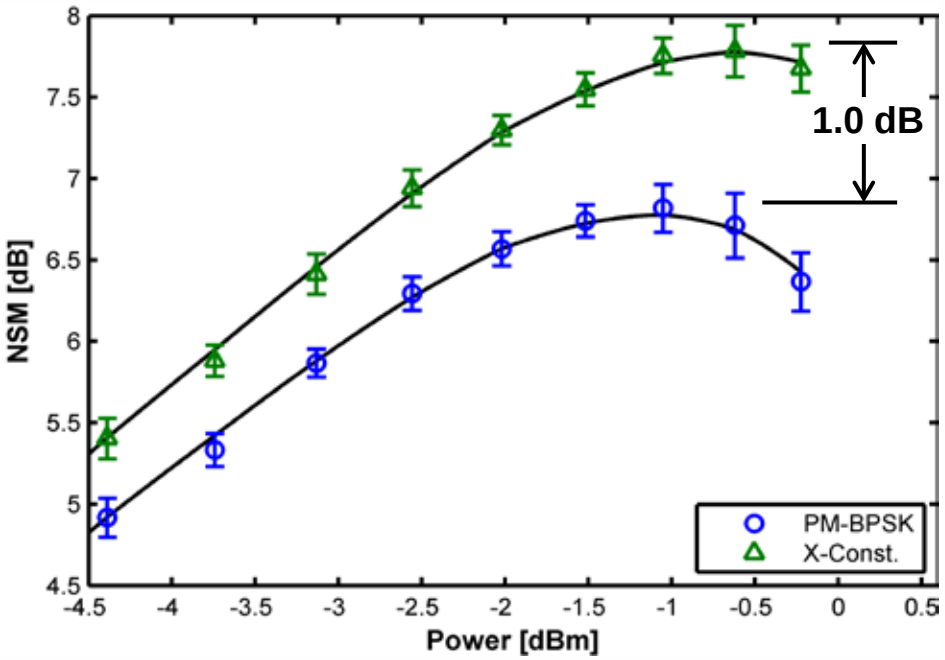
Experimental Verification



- Propagated 60 channels through a 5000 km (76 span) slope matched link.
- 37.5 GHz channel spacing for probe channel and four nearest neighbors on either side in a High Density (HD)-WDM configuration.
- 50 GHz spacing for the other 51 channels.
- 35 GSymb/s Tx waveforms generated with Ciena WL3 optical modems.
- Captured with a coherent receiver and processed offline.

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Experimental Results





Results comparing the X-Constellation to PM-BPSK following 5000 km of HD-WDM propagation:

- At low launch powers the X-Constellation's Euclidean distance gain results in a ~0.5 dB improvement in NSM
- At optimal launch power the X-Constellation increases NSM by 1.0 dB
- This corresponds to 2.2 dBQ at the link delivered OSNR of 14.1 dB

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Metro-optimized Coherent Evolution

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Economics

- Smaller footprint
- Lower power consumption
- Higher density

+

Technical

- Metro-optimized performance and latency
- Co-propagating 10G
- Network layer convergence for economical scale

High density 100G

- Lowest power dissipation, highest 100G port densities
- Retain key coherent benefits, optimize for metro
- Discrete for lowest cost; pluggable for multi-port

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WaveLogic Evolution:

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100G density and power improvements	• Smaller form-factor for higher faceplate density • Lower electrical power consumption
Integration of Tx and Rx DSP into a single ASIC	• Same Tx spectral shaping • Same Rx coherent detection DSP • Same PMD compensation: 150ps mean DGD • Same unmatched soft FEC; at lower power • Optimized modulation format for 100G metro/regional • Chromatic dispersion compensation scaled to metro/regional
Tx/Rx Electro-Optics	• New, smaller electro-optics

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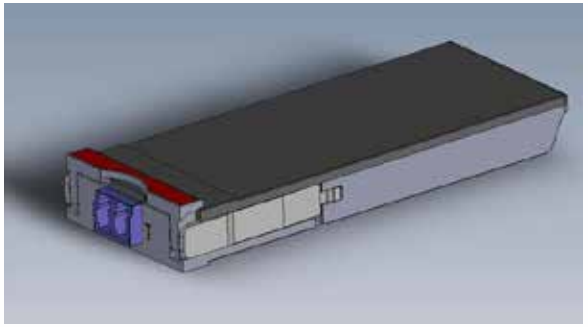
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Even higher density: CFP2-ACO

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OIF OPTICAL INTERNETWORKING FORUM

Ciena lead editor



Pluggable integrated optical component	• Tunable laser • Coherent optical modulator • Coherent receiver
Partitions optics and DSP	• Coherent DSP engines on the Host board; electro-optic conversion functions within pluggable **A**nalog **C**oherent **O**ptics module • Optimal thermal management for both optics and DSP • Allows innovation & competition across electro-optics technologies
Density	• Enables highest possible modem faceplate density for applications requiring more than two 100G coherent line ports
Pluggable	• **Pay as you grow**: addresses first-in install costs in multi-port line cards • **Flexibility**: allows selection of best-fit optical solution per application • **Reliability**: hot pluggable replacement in the event of failure
Supply chain	• Multiple vendors; large industry ecosystem

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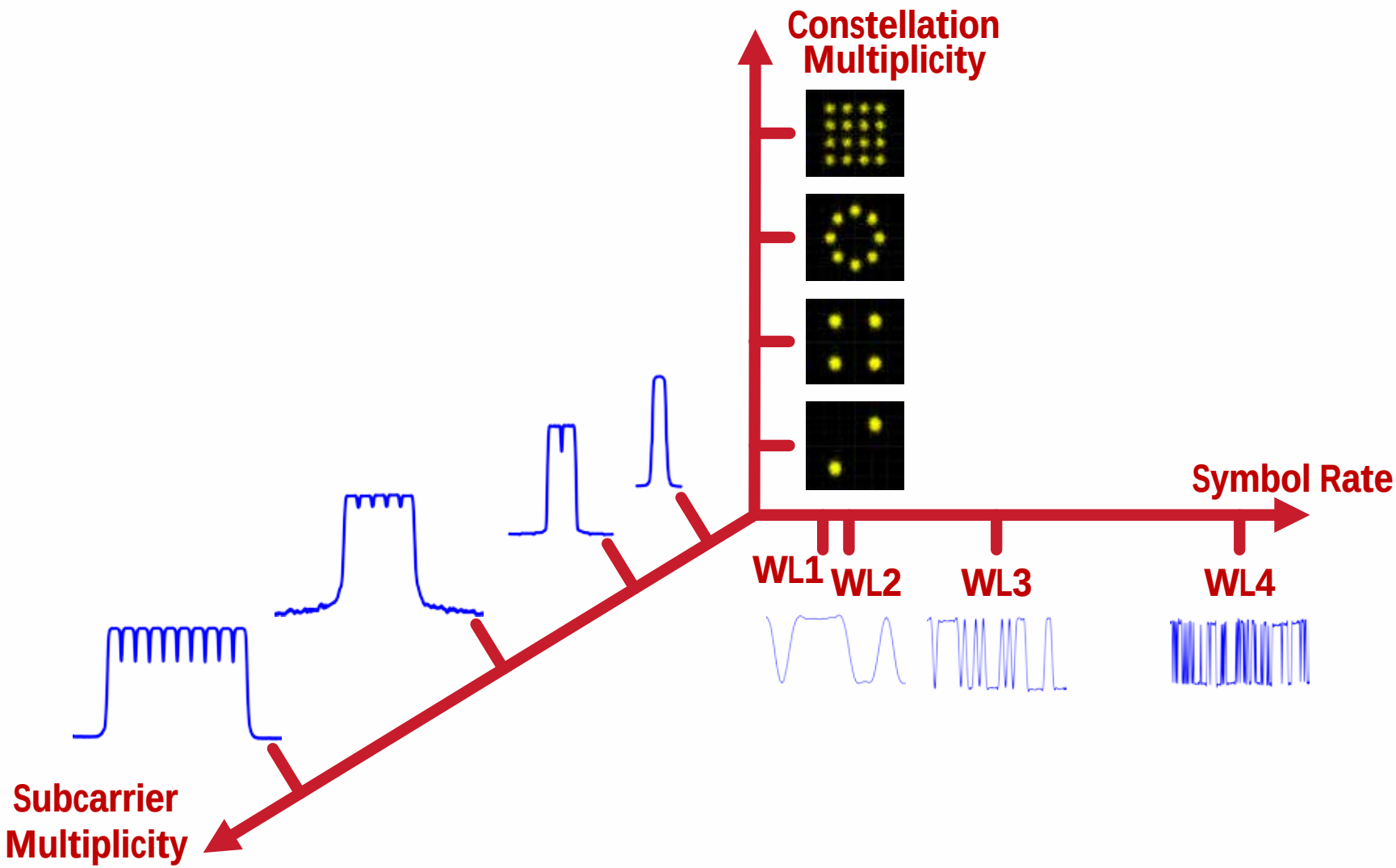
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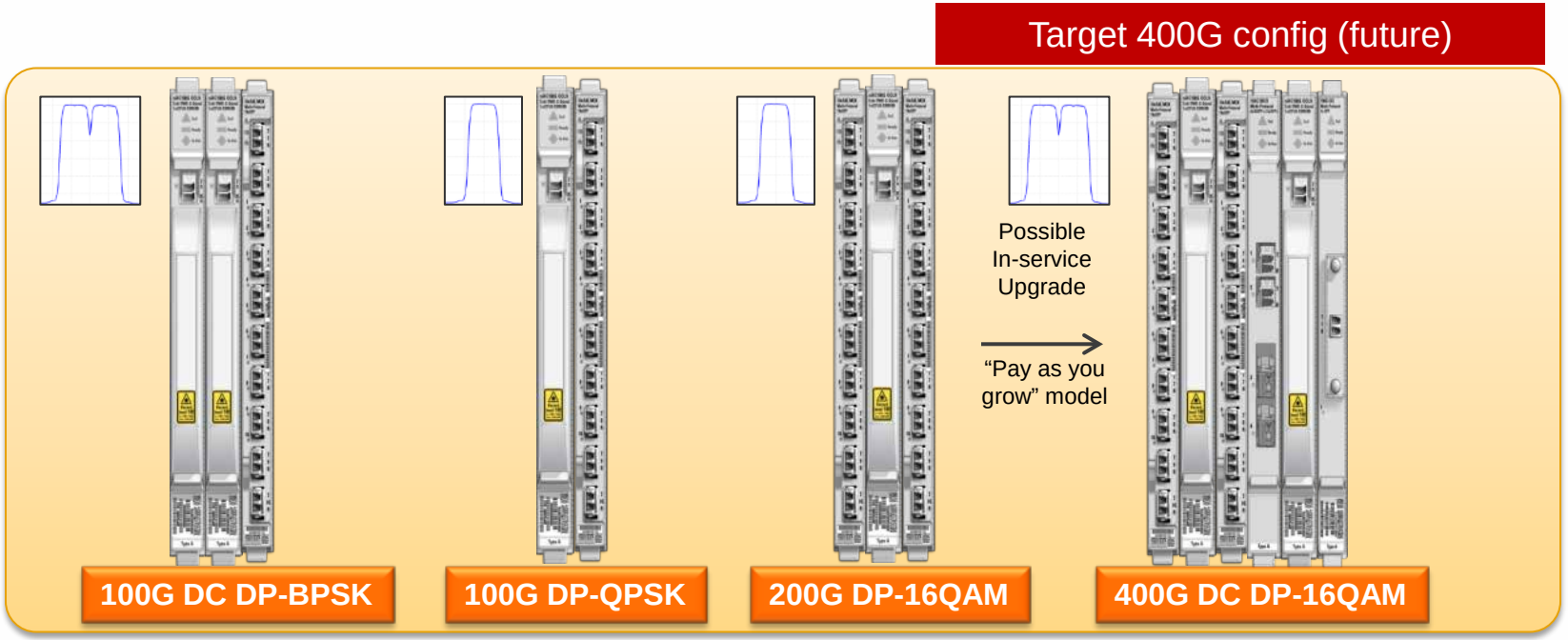
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Three Dimensions of Capacity Evolution



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Flex3 WaveLogic 3 Capabilities



Flex3 WaveLogic 3 can be deployed in various configurations to meet specific network needs:

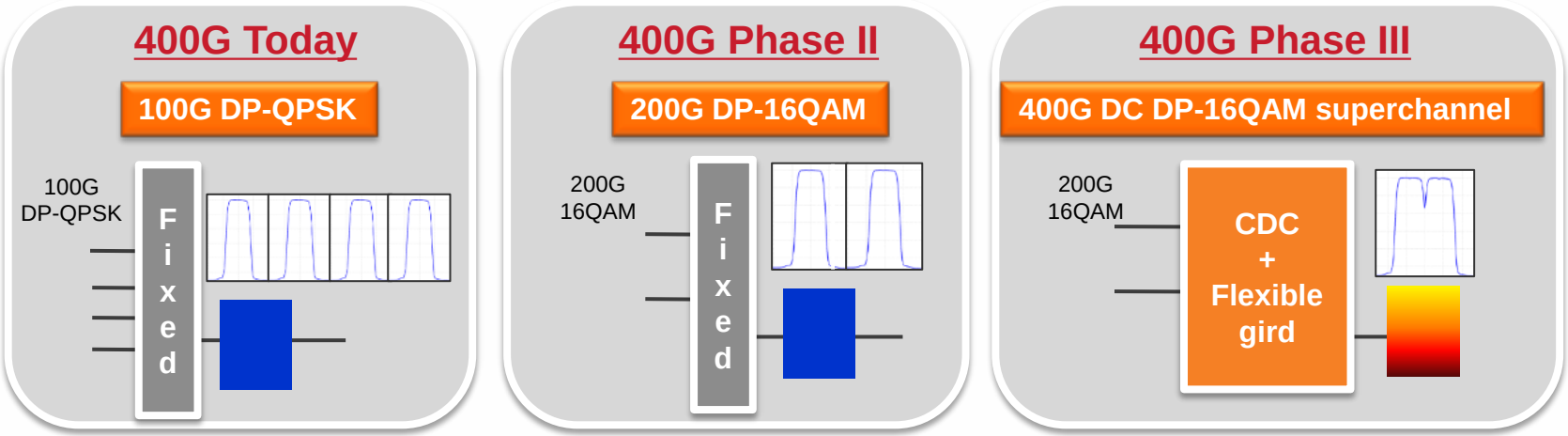
- supports multiple modulation formats for 100G client services
- is hardware-ready for future 400GbE transport solutions

Flex3 WaveLogic 3 200G (DP-16QAM) mode:

- allows independent mix-and-match of 100G client services (e.g. 10x10G MUX + 100G OCI)
- provides "pay-as-you-grow" from 100G→400G of traffic.

16QAM performance: 800-1000km without Raman, 1300km+ w/ Raman
Double the capacity over same spectrum in more locations à no Regens required

Scaling Network Capacity



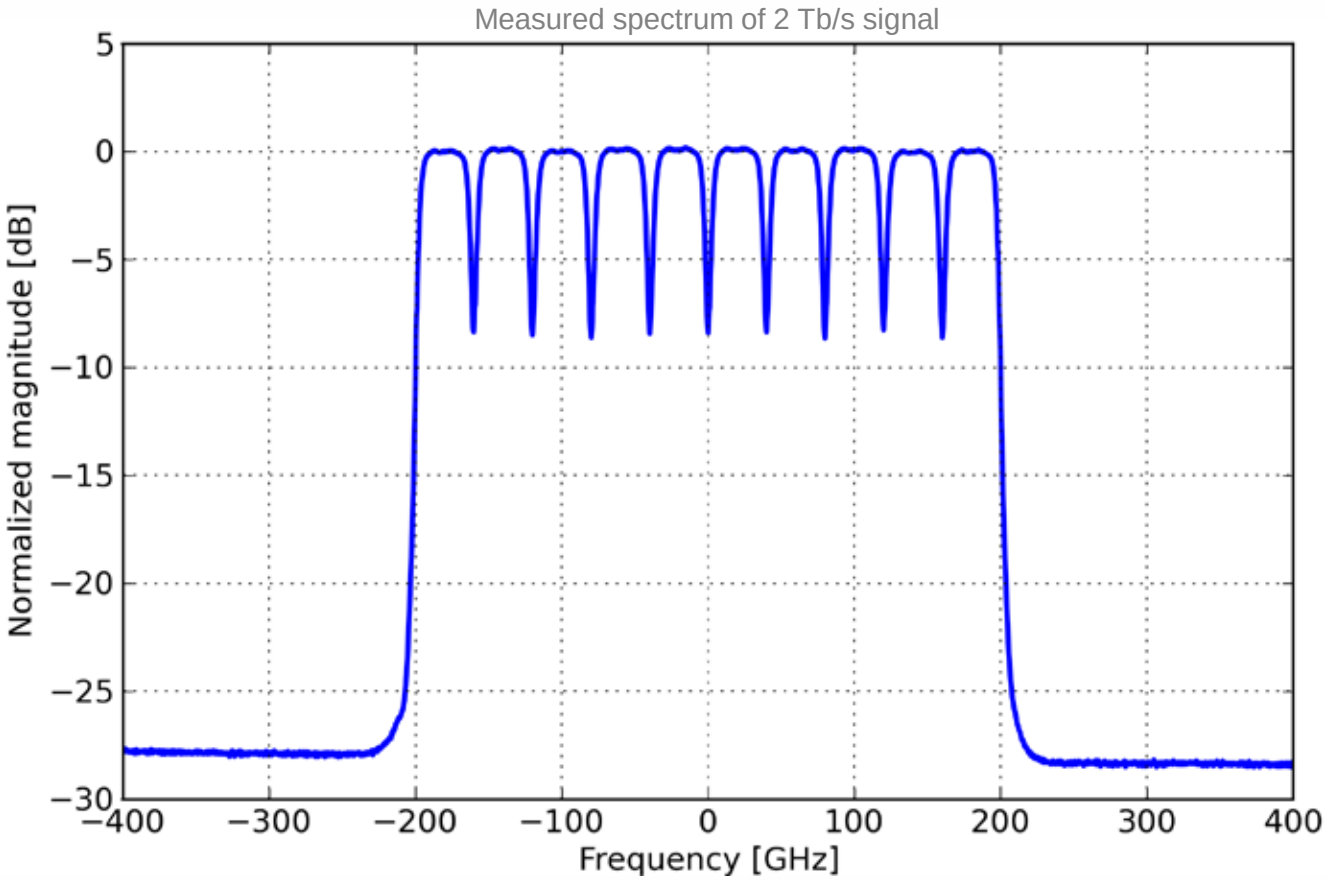
400G using 100G carriers	400G Phase 1	400G Phase II
88 x100G Waves @ 50GHz	88 x 200G Waves @ 50GHz	64 x 400G Superchannel @ 75GHz, flexible grid
4 channels for 400Gb/s	2 channels	Superchannel
8.8T, 200GHz, **SE 2.0 b/s/Hz**	**17.6T**, 100GHz, **SE 4.0 b/s/Hz**	**25.6T**, 75GHz , **SE 5.33 b/s/Hz**
> 3,000km	800-1400km	700-1300km

WaveLogic 3 200G (DP-16QAM) mode doubles capacity and spectral efficiency. Flexible grid tuning (future) allows for additional spectral efficiency improvements.

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2 Tb/s Super-Channel

- 10 subcarriers x 200 Gb/s (DP-16QAM)
- Df: 40 GHz
- Bandwidth occupancy: 400 GHz
- Spectral Efficiency: 5.0 b/s/Hz



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MPLS JAPAN

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